

```
; ModuleID = 'main.c'
```

```
source_filename = "main.c"
```

```
@.str = private unnamed_addr constant [4 x i8] c"%d\0A\00", align 1
```

```
; Function Attrs: noinline nounwind optnone uwtable
```

```
define i32 @fib(i32) #0 {
```

```
    %2 = alloca i32, align 4
```

```
    store i32 %0, i32* %2, align 4
```

```
    %3 = load i32, i32* %2, align 4
```

```
    %4 = icmp eq i32 %3, 0
```

```
    br i1 %4, label %5, label %6
```

```
5:
```

```
    ret i32 0
```

```
6:
```

```
    %7 = icmp eq i32 %3, 1
```

```
    br i1 %7, label %8, label %9
```

```
8:
```

```
    ret i32 1
```

```
9:
```

```
    %10 = load i32, i32* %2, align 4
```

```
    %11 = sub nsw i32 %10, 2
```

```
    %12 = call i32 @fib(i32 %11)
```

```
    %13 = load i32, i32* %2, align 4
```

```
    %14 = sub nsw i32 %13, 1
```

```
    %15 = call i32 @fib(i32 %14)
```

```
    %16 = add i32 %12, %15
```

```
    ret i32 %16
```

```
}
```

```
define i32 @main() #0 {
```

```
    %1 = alloca i32, align 4
```

```
    store i32 30, i32* %1, align 4
```

```
    %2 = load i32, i32* %1, align 4
```

```
    %3 = call i32 @fib(i32 %2)
```

```
    %4 = call i32 (i8*, ...) \
```

```
        @printf(i8* noundef getelementptr inbounds \
```

```
            ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef %3)
```

```
    ret i32 0
```

```
}
```

```
declare i32 @printf(i8* noundef, ...) #1
```